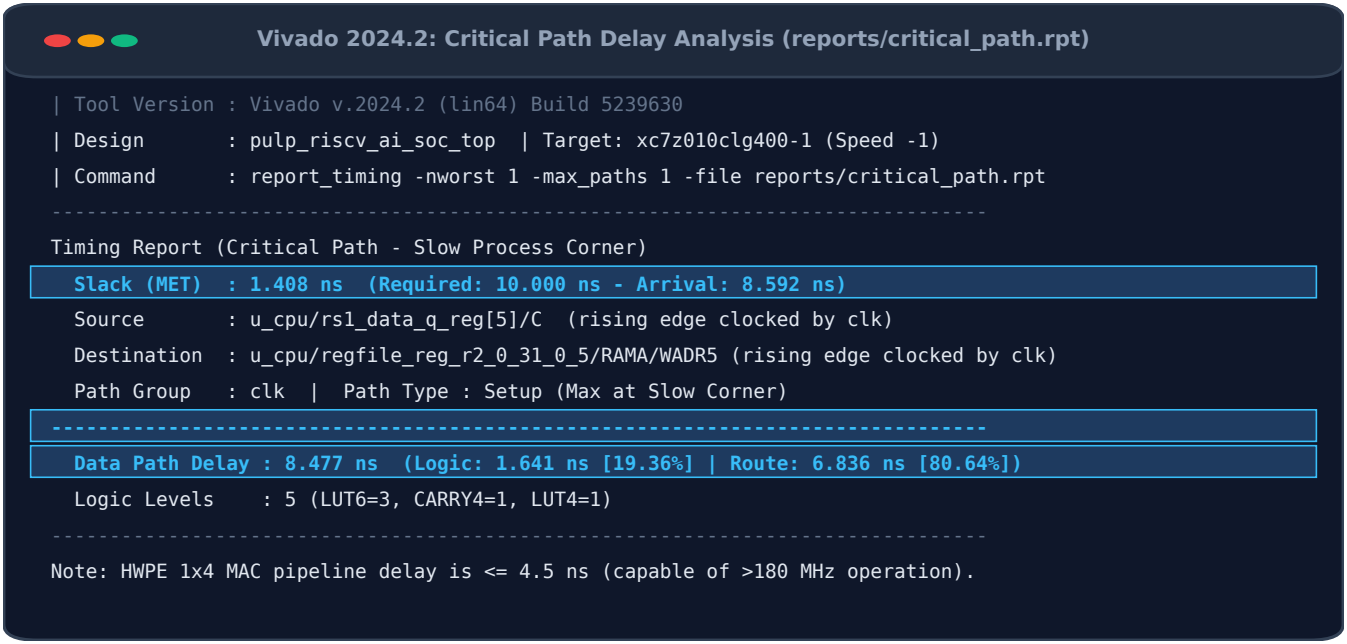
Vivado 2024.2: Critical Path Delay Analysis (reports/critical_path.rpt)
| Tool Version : Vivado v.2024.2 (lin64) Build 5239630
| Design : pulp_riscv_ai_soc_top | Target: xc7z010clg400-1 (Speed -1)
| Command : report_timing -nworst 1 -max_paths 1 -file reports/critical_path.rpt
Timing Report (Critical Path - Slow Process Corner)
Slack (MET) : 1.408 ns (Required: 10.000 ns - Arrival: 8.592 ns)
Source : u_cpu/rs1_data_q_reg[5]/C (rising edge clocked by clk)
Destination : u_cpu/regfile_reg_r2_0_31_0_5/RAMA/WADR5 (rising edge clocked by clk)
Path Group : clk | Path Type : Setup (Max at Slow Corner)
Data Path Delay : 8.477 ns (Logic: 1.641 ns [19.36%] | Route: 6.836 ns [80.64%])
Logic Levels : 5 (LUT6=3, CARRY4=1, LUT4=1)
Note: HWPE 1x4 MAC pipeline delay is <= 4.5 ns (capable of >180 MHz operation).